

01 THE PROBLEM

Small models fail on procedure
7B and 8B open-weight models, investigating alone

0.083

Qwen2.5-7B: share verified
picks its own probes, repeats them, runs out of budget
● ○ ○ ○ ○ ○ ○ ○ ○ ○ same probe, ten times

0 / 3,389

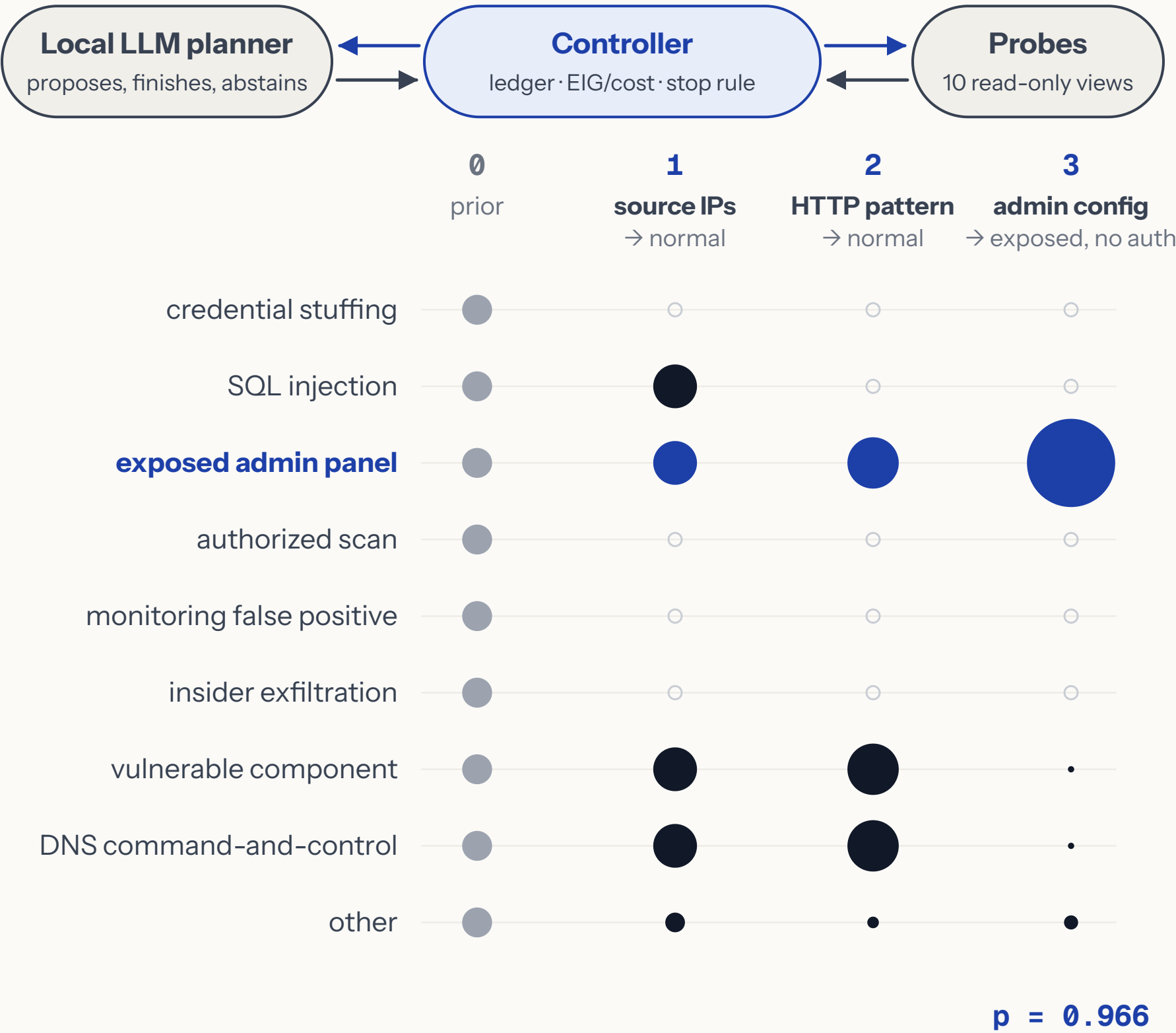
Llama-3.1-8B: verdicts / decisions
always asks for one more probe, even at $p > 0.999$
| | | | | | | | | | ○ no verdict

0.069

Qwen2.5-7B with its own tables
cannot write calibrated probe-outcome likelihoods
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ flat likelihoods

02 THE CONTROLLER

Hypotheses eliminated, probe by probe



The controller stops and cites evidence; the verifier confirms “exposed admin panel”.
Real episode sec-base-02, stopping study. Circle area = posterior; rings = eliminated.

03 WHAT WE MEASURED

Four pre-registered studies
5 models, 2 families, 7,272 audited episodes

- No oracle needed
0.125 → 1.000
Tables counted from LLM-free runs work as well as the true generator (Qwen 7B)
- The ranking helps every model
+0.26 to +0.35
EIG/cost vs random, same planner prompt; taking choice away: 7B +0.56, 72B -0.28
- What to probe ≠ when to stop
0 → 0.917
A controller stop rescues Llama 8B; probe choice adds nothing detectable after
- The controller alone comes close
0.917 vs 1.000
No LLM vs the best LLM configuration: strong models wait for confirming evidence
- Public SOC data cannot test this
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ExCyTIn, GUIDE and OTRF: the alert names the answer, or labels follow policy